Iodine Spectrometry

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Abstract

The visible absorption spectrum of molecular iodine was recorded using a tungsten source and a motor-driven monochromator. Mercury emission lines calibrated the wavelength scan, while a He–Ne laser provided an internal reference. The measured laser linewidth was $0.327\,\mathrm{nm}$, and the sodium doublet was resolved with a separation of $0.563\,\mathrm{nm}$. Birge–Sponer analysis focused on the progression assigned to $v'' = 0$, giving $\omega'_e = (127.1 \pm 1.1)\,\mathrm{cm}^{-1}$ and $\omega'_e x'_e = (1.0060 \pm 0.0089)\,\mathrm{cm}^{-1}$. The corresponding force constant was $(60.4 \pm 1.0)\,\mathrm{N\,m}^{-1}$, and Morse extrapolation gave a ground-state dissociation-energy estimate of $(145.43 \pm 0.39)\,\mathrm{kJ\,mol}^{-1}$. Repeated scans reproduced the principal spectral features, with harmonic constants differing from the primary result by at most 1.3%. Although the anharmonicity exceeded the literature values, it lay within published experimental ranges from comparable Birge–Sponer analyses using the same two-constant approximation. The measurements therefore support the usefulness of the method over the observed vibrational interval while illustrating the limitations of interpreting its fitted constants and extrapolated dissociation energies as a complete description of the molecular potential.

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1 Introduction

The visible absorption spectrum of molecular iodine provides a direct means of investigating molecular vibrational structure [1, 2]. Electronic excitation is accompanied by changes in vibrational and rotational energy, producing a series of absorption bands. Their positions and decreasing separations contain information about the molecular potential and its departure from harmonic behaviour [3].

Birge–Sponer analysis relates successive vibrational spacings to vibrational constants and dissociation energies [1, 4]. In this experiment, visible iodine spectra were used to determine the excited-state harmonic constant, anharmonicity and force constant. Extrapolation of the fitted vibrational structure then provided estimates of the excited- and ground-state dissociation energies.

The recording labelled WIn4903 was used for the primary analysis, while two additional recordings assessed repeatability. The analysis focused on the progression assigned to $v'' = 0$. The $v'' = 1$ progression provided a comparison over a different vibrational interval and a test of sensitivity to overlapping spectral features.

2 Theory

2.1 Absorption and vibrational structure

The visible iodine spectrum includes the transition $B\,^3\Pi_{0u}^+ \leftarrow X\,^1\Sigma_g^+$. Primes and double primes denote upper- and lower-state vibrational quantum numbers, respectively. Progressions originating from different lower vibrational levels therefore probe the same upper electronic state.

The apparent absorbance and band wavenumber are

$$A(\lambda) = \log_{10}\left[\frac{I_0(\lambda)}{I(\lambda)}\right], \tag{1}$$

$$\widetilde{\nu}\ (\mathrm{cm}^{-1}) = \frac{10^7}{\lambda\ (\mathrm{nm})}. \tag{2}$$

Here, I_0 and I are the reference and transmitted intensities at corresponding wavelengths. Detector voltages are proportional to these intensities for a linear response. Without a separate dark correction, their ratio gives an apparent absorbance.

To second order, the vibrational term value is

$$G(v) = \omega_e\left(v + \frac{1}{2}\right) - \omega_e x_e\left(v + \frac{1}{2}\right)^2. \tag{3}$$

For adjacent bands sharing the same lower vibrational level, the lower-state contribution cancels:

$$\Delta G_B = \widetilde{\nu}_{v'+1,v''} - \widetilde{\nu}_{v',v''} = \omega_e' - 2\omega_e' x_e'(v' + 1). \tag{4}$$

The Birge–Sponer fit is expressed using the midpoint of the two upper-state quantum numbers, $x = v' + \frac{1}{2}$:

$$\Delta G_B = C + Mx, \qquad \omega_e' x_e' = -\frac{M}{2}, \qquad \omega_e' = C - \frac{M}{2}. \tag{5}$$

Consequently, the intercept is $C = \omega_e' - \omega_e' x_e'$, rather than ω_e' itself. Absolute vibrational numbering is required to interpret the intercept.

At the experimental resolution, the measured maxima represent unresolved band structure and need not coincide exactly with rotation-free band origins.

2.2 Force constant and dissociation energies

The excited-state force constant is

$$k_B = \mu(2\pi c\,100\omega'_e)^2, \qquad \mu = \frac{m_{\mathrm{I}}}{2}, \tag{6}$$

where c is in $\mathrm{m\,s^{-1}}$, μ is in kilograms, and ω'_e is in $\mathrm{cm^{-1}}$. The factor of 100 converts the spectroscopic wavenumber to $\mathrm{m^{-1}}$.

Writing $\widetilde{D} = D/(hc)$, the Morse approximation gives

$$\widetilde{D}_{e,B} = \frac{(\omega'_e)^2}{4\omega'_e x'_e}, \qquad \widetilde{D}_{0,B} = \widetilde{D}_{e,B} - \frac{\omega'_e}{2} + \frac{\omega'_e x'_e}{4}. \tag{7}$$

The subscripts e and 0 refer to dissociation from the potential minimum and the lowest vibrational level, respectively.

For an assigned reference band originating from $v'' = 0$, the convergence energy and ground-state dissociation energies follow from

$$\widetilde{E}^{*} = \widetilde{\nu}_{v',0} + \widetilde{D}_{e,B} - G_B(v'), \tag{8}$$

$$\widetilde{D}_{0,X} = \widetilde{E}^{*} - \widetilde{E}_{\mathrm{SO}}, \tag{9}$$

$$\widetilde{D}_{e,X} = \widetilde{D}_{0,X} + \frac{\omega''_e}{2} - \frac{\omega''_e x''_e}{4}. \tag{10}$$

The atomic fine-structure separation is $\widetilde{E}_{\mathrm{SO}} = 7603.15\,\mathrm{cm^{-1}}$ [7]. The ground-state constants $\omega''_e = 214.502\,\mathrm{cm^{-1}}$ and $\omega''_e x''_e = 0.6147\,\mathrm{cm^{-1}}$ give a zero-point correction of $107.10\,\mathrm{cm^{-1}}$ [6]. Spectral energies are converted to molar energies using $N_{\mathrm{A}}hc$, with consistent length units.

These dissociation estimates depend on extrapolating the two-constant vibrational model beyond the observed levels.

3 Experimental procedure and apparatus

3.1 Optical arrangement and acquisition

A tungsten lamp provided a continuous source. Spectra were recorded with and without the iodine cell using a scanning monochromator and photomultiplier detector **(Fig. 1)**. Reference and iodine signals were compared at corresponding wavelengths using Eq. (1).

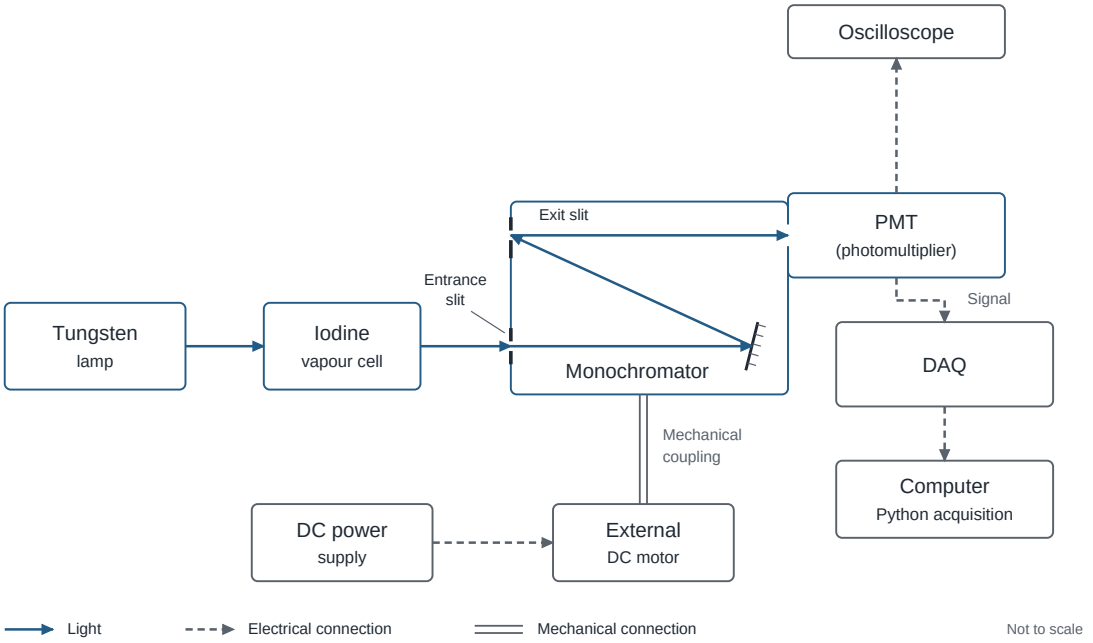


Fig. 1. Optical arrangement and signal acquisition used to record the iodine absorption spectrum.

Detector readings were acquired in blocks of 20 samples at a nominal rate of 10 000 Hz. Each block was averaged to produce one stored value, with a programmed delay of 0.001 s between acquisitions. The stored sampling interval also included acquisition and software overhead; the wavelength calibration therefore used the recorded times.

Direct illumination of the exit slit was checked by comparing the oscilloscope signal with the light source removed and with the entrance slit covered. These checks were repeated throughout the experiment.

3.2 Resolution and wavelength calibration

Resolution was assessed using the sodium doublet and the He–Ne laser profile. Temporal linewidths were converted to wavelength, and the effective resolving power was calculated using

$$\mathrm{FWHM}_\lambda = |b|\,\mathrm{FWHM}_t, \qquad R = \frac{\lambda_L}{\mathrm{FWHM}_{\lambda,L}}. \tag{11}$$

The measured laser profile characterises the response of the complete scanning measurement.

An external DC motor drove the monochromator. The supply was operated in constant-voltage mode, with the current limit above the operating demand. Mercury emission lines calibrated the wavelength–time relationship:

$$\lambda(t) = a + bt. \tag{12}$$

Forward and reverse scans were fitted separately. Repeated calibrations were compared using common mercury lines, and residuals were inspected for departures from linearity.

The He–Ne line at nominal wavelength $\lambda_L = 632.8\,\mathrm{nm}$ provided an internal reference for each iodine and tungsten recording:

$$\lambda(t) = \lambda_L + b(t - t_L), \tag{13}$$

where t_L is the measured laser-line position. This removed the offset associated with manual start timing while retaining the mercury-derived scan speed.

3.3 Band selection

Peaks were detected in smoothed apparent-absorbance spectra and assigned to progressions by inspection. The reference band in the $v'' = 0$ progression was assigned $v' = 29$. For the $v'' = 1$ progression, the selected band nearest the reference wavelength 581.278 nm was assigned $v' = 15$. Other selected bands were numbered consecutively in increasing wavenumber. The WIn4903 fits used 30 bands in the $v'' = 0$ sequence and 19 in the $v'' = 1$ sequence; the latter supplied 17 retained spacings after the stated exclusion.

The $v'' = 1$ spacing at midpoint 14.5 was excluded from each fit following inspection of the overlapping spectral features in that region. The excluded spacing was retained on the plots. The same assignments and exclusion were applied to all three recordings.

4 Uncertainty treatment

The wavelength uncertainty included contributions from the mercury-derived scan speed, iodine-band location and He–Ne reference position. Half the measured laser FWHM was adopted as the reference-position uncertainty:

$$u_L = \frac{\mathrm{FWHM}_{\lambda,L}}{2} \simeq 0.164\,\mathrm{nm}. \tag{14}$$

The unrounded half-width was retained in the WIn4903 calculation. This is an assigned linewidth-based position allowance, rather than uncertainty in the known laser wavelength or a statistical estimate of the fitted line centre.

For independent speed, band-time and reference-position contributions, first-order propagation gives

$$u_{\lambda_i}^2 = (t_i - t_L)^2 u_b^2 + b^2 u_{t_i}^2 + u_L^2, \qquad u_{\widetilde{\nu}_i} = \frac{10^7}{\lambda_i^2} u_{\lambda_i}. \tag{15}$$

The laser contribution is included once as a shared wavelength offset. The adopted speed uncertainty accounts for variation in transferring the mercury calibration to an individual iodine scan.

The shared calibration contributions produce correlations between band positions. Defining $g_i = -10^7(t_i - t_L)/\lambda_i^2$ and $h_i = -10^7/\lambda_i^2$, the covariance is

$$\mathbf{C}_\nu = \mathrm{diag}(u_{\mathrm{peak},i}^2) + u_b^2 \mathbf{g}\mathbf{g}^\mathsf{T} + u_L^2 \mathbf{h}\mathbf{h}^\mathsf{T}. \tag{16}$$

Here, $u_{\mathrm{peak},i}$ is the independent band-location contribution in cm^{-1}. It was estimated from scatter about a quadratic progression, with a sampling-based lower bound, and is therefore a model-based uncertainty estimate.

A difference matrix $\mathbf{D}$ propagated the band-position covariance to adjacent spacings:

$$\mathbf{C}_{\Delta G} = \mathbf{D}\mathbf{C}_\nu\mathbf{D}^\mathsf{T}. \tag{17}$$

This retains both the shared calibration contributions and correlations between neighbouring spacings that contain a common band position.

Generalised least squares was used for the Birge–Sponer fits. Derived uncertainties were obtained from the fitted parameter covariance:

$$u_f^2 = \boldsymbol{\nabla} f^\mathsf{T} \mathbf{C}_\theta \boldsymbol{\nabla} f. \tag{18}$$

In particular, the intercept–slope covariance was retained when calculating the harmonic constant, force constant and dissociation energies. For the convergence and ground-state energies, the covariance between the selected reference-band position and the fitted parameters was also

retained. The external atomic and ground-state constants were treated as fixed. The resulting uncertainties remain conditional on the adopted model.

The quoted uncertainties describe the adopted measurement and fitting contributions. They do not include uncertainty from vibrational assignment, systematic scan nonlinearity or extrapolation of the Morse approximation.

5 Results

5.1 Resolution and calibration

The sodium doublet was resolved into two distinct maxima, with a measured separation of 0.563 nm compared with the reference value of 0.5974 nm. The measured He–Ne linewidth was 0.327 nm. Using Eq. (11), this corresponds to an effective resolving power of 1.93×10^3. The corresponding spectra are shown in Appendix B.

The mercury calibration gave forward and reverse scan speeds of $(1.833 \pm 0.012)\,\mathrm{nm\,s^{-1}}$ and $(-1.8580 \pm 0.0088)\,\mathrm{nm\,s^{-1}}$, respectively. The wavelength–time relationships were approximately linear, with residual departures of order a few tenths of a nanometre **(Fig. 2)**.

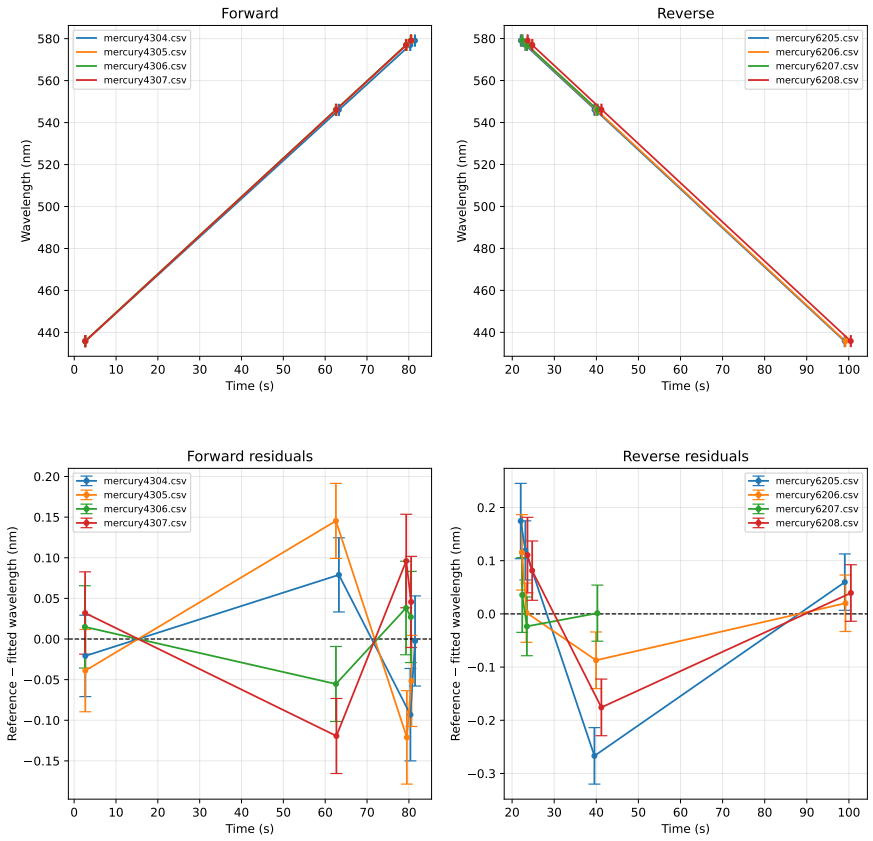


Fig. 2. Mercury wavelength calibration for forward and reverse scans. Upper panels show the fitted wavelength–time relationships; lower panels show reference minus fitted wavelengths. Error bars represent the adopted timing allowances, converted to wavelength units in the residual plots.

5.2 Absorption spectrum

The three iodine recordings reproduced the principal absorption features despite differences in baseline and amplitude **(Fig. 3)**. Taking the reference-to-sample ratio reduced the broad source and detector response, making the vibrational structure clearer than in the direct voltage difference.

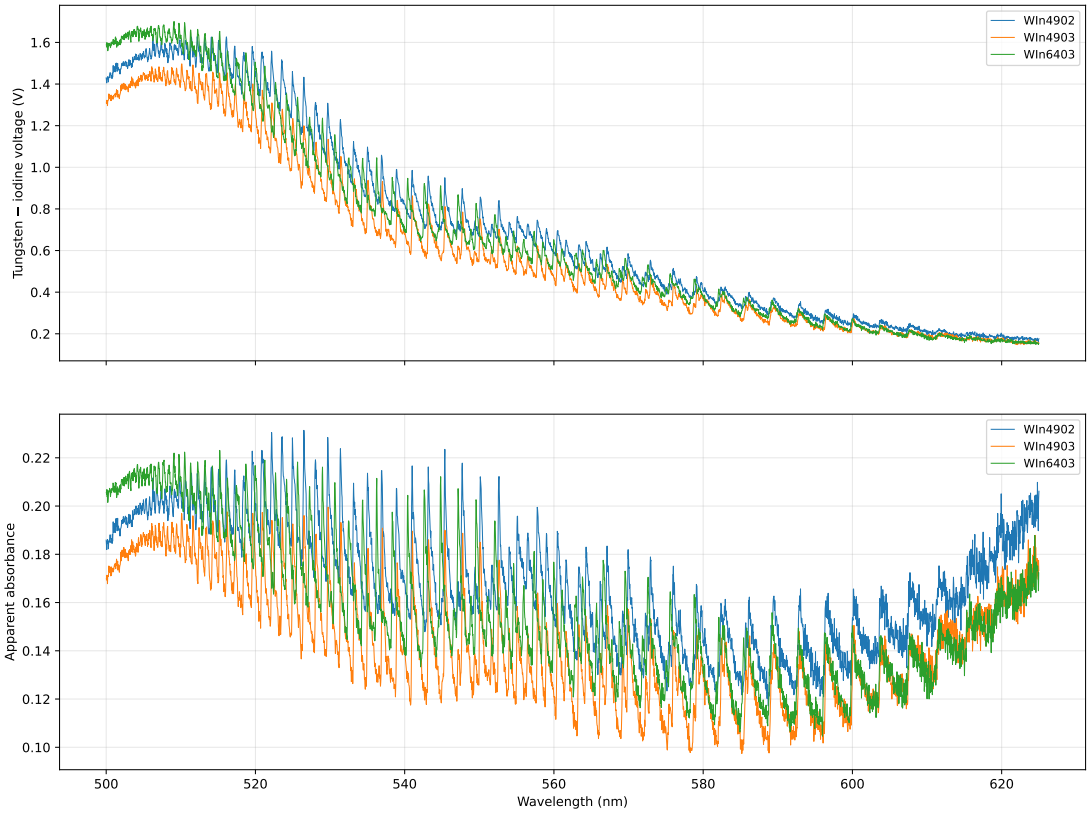


Fig. 3. Comparison of three iodine recordings. Upper panel: tungsten reference minus iodine detector signal. Lower panel: apparent absorbance used for subsequent band identification.

The selected WIn4903 bands were compared with tabulated reference positions in **(Fig. 4)**. The progressions assigned to $v'' = 0$ and $v'' = 1$ were analysed independently.

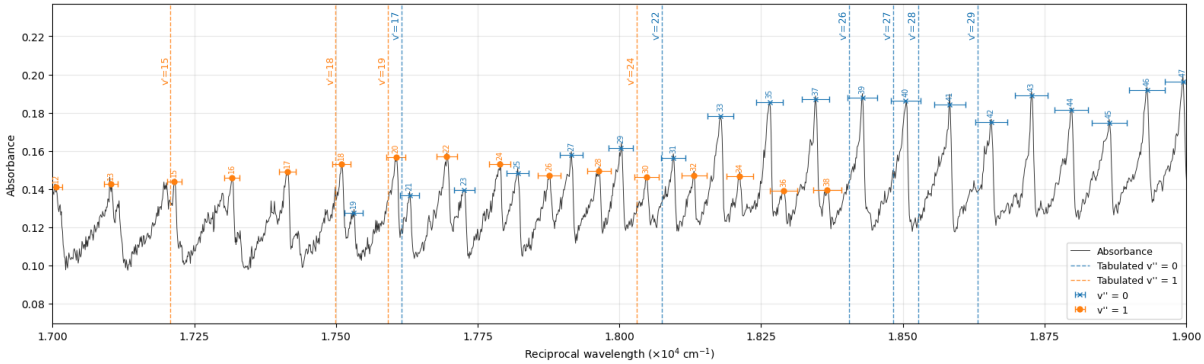


Fig. 4. Band assignment in WIn4903. Blue crosses and orange circles identify the progressions assigned to $v'' = 0$ and $v'' = 1$, respectively. Dashed vertical lines indicate tabulated transition positions. Horizontal error bars show the propagated mercury scan-speed contribution.

The anomalous interval between the bands assigned to $v' = 14$ and $v' = 15$ in the $v'' = 1$ progression coincides with overlapping spectral features. A change in the component followed by the peak-selection procedure is a plausible explanation for this interval. It was excluded from the fit as described in Section 3; the remaining selected bands were retained.

5.3 Birge–Sponer analysis

The fitted intercepts and slopes are listed in Table 1, using the midpoint convention in Eq. (5). The corresponding fits are shown in **(Fig. 5)**.

Table 1. WIn4903 Birge–Sponer fit parameters. Both C and M are expressed in cm^{-1}; the vibrational coordinate is dimensionless.

Progression	Intercept C	Slope M
$v'' = 0$	126.1 ± 1.1	-2.012 ± 0.018
$v'' = 1$	118.2 ± 1.3	-1.500 ± 0.066

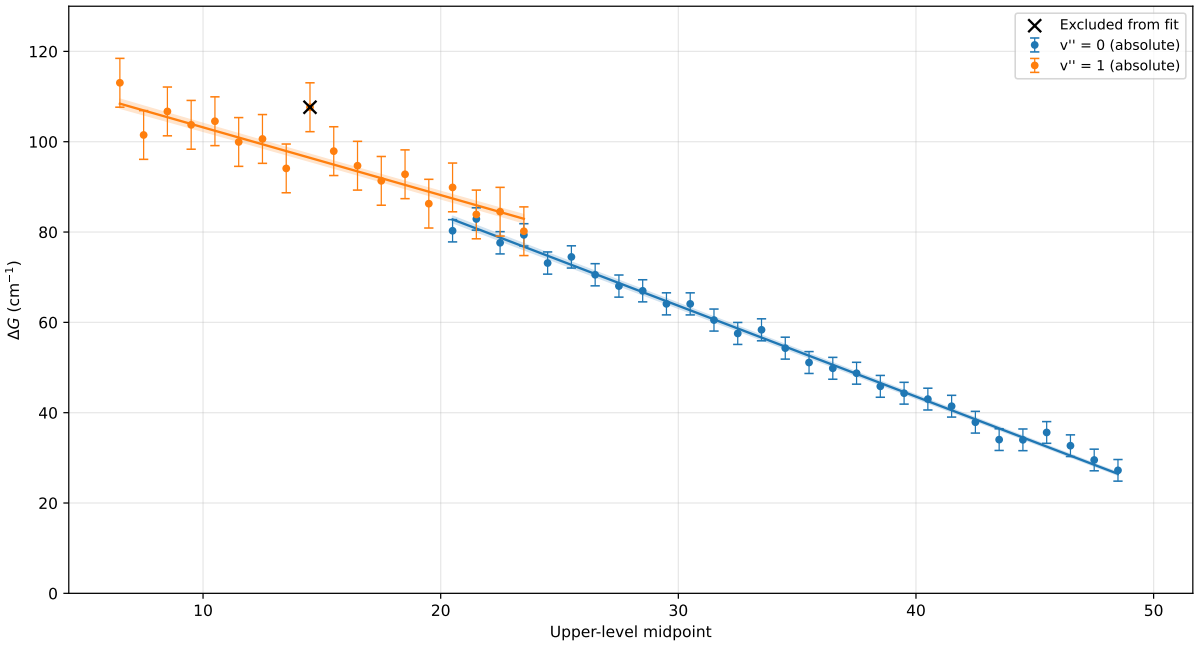


Fig. 5. Birge–Sponer plots for the $v'' = 0$ and $v'' = 1$ progressions in WIn4903. Error bars show propagated spacing uncertainties, and shaded regions indicate uncertainty in the fitted mean. The $v'' = 1$ spacing at midpoint 14.5 is displayed but excluded from the fit.

Using Eq. (5), the $v'' = 0$ progression gives $\omega'_e = (127.1 \pm 1.1)\,\mathrm{cm}^{-1}$ and $\omega'_e x'_e = (1.0060 \pm 0.0089)\,\mathrm{cm}^{-1}$. The force constant calculated using Eq. (6) is $k_B = (60.4 \pm 1.0)\,\mathrm{N\,m}^{-1}$.

For the $v'' = 1$ progression, the same procedure gives $\omega'_e = (118.9 \pm 1.3)\,\mathrm{cm}^{-1}$ and $\omega'_e x'_e = (0.750 \pm 0.033)\,\mathrm{cm}^{-1}$, corresponding to $k_B = (52.9 \pm 1.2)\,\mathrm{N\,m}^{-1}$.

Table 2. Primary WIn4903 constants and the typical experimental ranges reported by Lessinger for linear Birge–Sponer analyses [4]. All values are in cm^{-1}. These ranges describe student measurements, not uncertainty intervals on accepted constants.

Quantity	WIn4903, $v'' = 0$	Lessinger's range
ω'_e	127.1 ± 1.1	127–135
$\omega'_e x'_e$	1.0060 ± 0.0089	0.94–1.05

5.4 Dissociation energies and literature comparison

The excited-state dissociation energies were calculated using Eq. (7). For $v'' = 0$, the results are $\widetilde{D}_{e,B} = (4012 \pm 35)\,\mathrm{cm}^{-1}$ and $\widetilde{D}_{0,B} = (3949 \pm 34)\,\mathrm{cm}^{-1}$. The $v'' = 1$ progression gives $\widetilde{D}_{e,B} = (4.72 \pm 0.13) \times 10^3\,\mathrm{cm}^{-1}$ and $\widetilde{D}_{0,B} = (4.66 \pm 0.13) \times 10^3\,\mathrm{cm}^{-1}$.

Table 3 compares the results with the historical reference values tabulated by Stafford [1]. Percentage difference denotes $100|Q - Q_{\mathrm{ref}}|/|Q_{\mathrm{ref}}|$, while relative uncertainty denotes $100u_Q/|Q|$. Percentages were calculated before rounding. The dependence of this comparison on the fitted vibrational model is considered in Section 6.

Table 3. WIn4903 results compared with the reference values tabulated by Stafford [1]. Vibrational constants are in cm^{-1}, force constants in $\mathrm{N\,m}^{-1}$, and dissociation energies in $10^3\,\mathrm{cm}^{-1}$.

Quantity	v''	Experimental	Reference	Difference (%)	Relative uncertainty (%)
ω'_e	0	127.1 ± 1.1	128	0.74	0.85
$\omega'_e x'_e$	0	1.0060 ± 0.0089	0.834	20.6	0.89
k_B	0	60.4 ± 1.0	61.3	1.5	1.7
$\widetilde{D}_{e,B}$	0	4.012 ± 0.035	4.398	8.8	0.87
$\widetilde{D}_{0,B}$	0	3.949 ± 0.034	4.335	8.9	0.87
ω'_e	1	118.9 ± 1.3	128	7.1	1.1
$\omega'_e x'_e$	1	0.750 ± 0.033	0.834	10.1	4.4
k_B	1	52.9 ± 1.2	61.3	13.7	2.2
$\widetilde{D}_{e,B}$	1	4.72 ± 0.13	4.398	7.2	2.9
$\widetilde{D}_{0,B}$	1	4.66 ± 0.13	4.335	7.4	2.9

The reference force constant was calculated from the reference harmonic constant using Eq. (6). Derived uncertainties retain the fitted-parameter covariance and exclude Morse-extrapolation error.

The WIn4903 band assigned to $(v', v'') = (29, 0)$ occurs at approximately 537.02 nm, corresponding to $18\,621\,\mathrm{cm}^{-1}$ using Eq. (2). Combining its unrounded position with the $v'' = 0$ fitted constants in Eq. (8) gives $\widetilde{E}^* \simeq 19\,760\,\mathrm{cm}^{-1}$.

Using Eq. (9) and Eq. (10) then gives the ground-state dissociation estimates in Table 4. The molar dissociation energy from the lowest vibrational level is approximately $(145.43 \pm 0.39)\,\mathrm{kJ\,mol}^{-1}$.

Table 4. Convergence and ground-state dissociation estimates from the WIn4903 $v'' = 0$ progression. All spectral energies are in cm^{-1}. Reference values are those tabulated by Stafford [1].

Quantity	Experimental	Reference	Difference (%)
$\widetilde{E}^*$	$19\,760 \pm 33$	20 040	1.4
$\widetilde{D}_{0,X}$	$12\,157 \pm 33$	12 440	2.3
$\widetilde{D}_{e,X}$	$12\,264 \pm 33$	12 550	2.3

Uncertainties retain the covariance between the reference-band position and the fitted constants. They exclude assignment, calibration-model and extrapolation errors; the external atomic and ground-state constants are treated as fixed.

Relative to WIn4903, the $v'' = 0$ harmonic constants from WIn4902 and WIn6403 differ by approximately -0.38% and $+1.31\%$, respectively. The individual scan results are given in Appendix A.

6 Discussion

The experiment resolved iodine vibrational structure and reproduced the principal absorption features in three recordings. The harmonic constants from the additional scans differed from the primary WIn4903 result by at most 1.3%. This repeatability supports the measurement procedure, although shared calibration and assignment assumptions are not tested by repetition alone.

For the principal $v'' = 0$ progression, ω'_e is 0.74% below Stafford's reference value, whereas $\omega'_e x'_e$ is 20.6% higher. The latter difference exceeds the propagated uncertainty, but should not be equated directly with a measurement error of the same magnitude. Both constants fall within Lessinger's typical experimental ranges for linear Birge–Sponer analyses (Table 2). Those ranges establish a precedent for the present results, rather than an independent validation of accuracy.

McNaught demonstrates the model dependence directly: including a cubic vibrational term changes the lower-order constants obtained from the same iodine measurements and brings them closer to tabulated values [5]. A linear fit to level spacings over a restricted vibrational interval consequently yields effective constants for that interval. Points distributed on both sides of a straight line do not establish that its coefficients must equal those of a higher-order expansion. For comparison, the NIST compilation gives $\omega'_e = 125.697\,\mathrm{cm}^{-1}$ and $\omega'_e x'_e = 0.7642\,\mathrm{cm}^{-1}$ within a higher-order description [6]; the present constants are approximately 1.1% and 31.6% higher, respectively.

The two progressions probe the same excited electronic state. Their different fitted constants therefore do not imply different physical force constants. They sample different vibrational intervals, and the $v'' = 1$ selection is additionally sensitive to overlapping features. Near the interval assigned to $v' = 14$ and $v' = 15$, the dominant maximum appears to switch between neighbouring spectral paths **(Fig. 4)**. This motivates excluding the spacing at midpoint 14.5, while retaining the subsequent points. A crude assumption was made that the adjacent-band spacings along the two paths are approximately equal near and beyond the crossover, so that switching paths principally affects the crossover interval itself. The identity of the neighbouring path was not established independently, and this assumption is not included in the propagated uncertainty.

Separate fits below and above the excluded interval give slopes of $(-1.59 \pm 0.43)\,\mathrm{cm}^{-1}$ and $(-1.91 \pm 0.33)\,\mathrm{cm}^{-1}$, respectively. Their difference is $(-0.31 \pm 0.54)\,\mathrm{cm}^{-1}$ when their shared covariance is retained. Thus, the apparent change in slope is not resolved by these data. Improved resolution and comparison with an assigned line list would help distinguish the $v'' = 1$ bands from the neighbouring progression. Following both paths across several bands and checking their combination differences would test the crossover assumption more directly than excluding a single interval.

The calibration also has finite limitations. Mercury residuals test departures from the assumed linear wavelength–time relation **(Fig. 2)**; these departures are distinct from variation in the mean scan speed. Resolving the sodium doublet demonstrates resolving capability, but its measured separation is 5.8% below the reference and does not establish exact wavelength accuracy. Calibrations immediately before and after each iodine recording, or independent monitoring of monochromator position, would better distinguish calibration effects from model dependence.

Extrapolation places a stronger demand on the model than fitting the observed bands. Lessinger attributes errors in linear Birge–Sponer dissociation estimates principally to departures from linear spacings near dissociation, where the Morse potential inadequately describes the long-range interaction [4]. The present excited-state estimates are 8.8–8.9% below Stafford's reference energies, compared with propagated uncertainties of approximately 0.87%. This underestimate is consistent with the elevated fitted anharmonicity: through Eq. (7), a larger $\omega'_e x'_e$ reduces the calculated well depth at fixed ω'_e. The anharmonicity and dissociation-energy discrepancies are therefore linked through the model, rather than independent indicators of experimental error. Their percentage differences need not coincide, since Stafford's reference dissociation energies are

not obtained by substituting his tabulated vibrational constants into the same Morse expression.

The ground-state estimate, $(145.43 \pm 0.39)\,\mathrm{kJ\,mol^{-1}}$, is 2.3% below the selected reference. Its smaller percentage difference does not independently validate the extrapolated potential: the estimate combines the extrapolation with the observed reference-band energy and atomic fine-structure separation. Its quoted uncertainty describes propagation within that model and does not include the extrapolation error.

The reproducible spectra and comparison with published Birge–Sponer results support the effectiveness of the experiment. The larger anharmonicity discrepancy is consistent with a limitation of the two-constant description, while the calibration and overlapping-band evidence prevent assigning the entire discrepancy uniquely to the model.

7 Conclusion

The calibrated scanning experiment successfully resolved reproducible vibrational progressions in molecular iodine. Analysis of the WIn4903 $v'' = 0$ progression gave $\omega'_e = (127.1 \pm 1.1)\,\mathrm{cm^{-1}}$, $\omega'_e x'_e = (1.0060 \pm 0.0089)\,\mathrm{cm^{-1}}$ and $k_B = (60.4 \pm 1.0)\,\mathrm{N\,m^{-1}}$. The principal harmonic constant was reproduced within 1.3% across the additional scans.

Although the anharmonicity exceeds the selected historical reference, both vibrational constants fall within the typical experimental ranges published for linear Birge–Sponer analyses. The measurements therefore provide a credible description of the observed vibrational interval and demonstrate the importance of comparing parameters obtained using compatible models.

Morse extrapolation gave a ground-state dissociation energy of approximately $(145.43 \pm 0.39)\,\mathrm{kJ\,mol^{-1}}$, 2.3% below the selected reference. The experiment illustrates both the effectiveness of calibrated absorption spectroscopy and the limits of extending a locally successful vibrational model to the dissociation threshold.

References

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A Repeatability

All three scans were analysed using the same assignment conventions and fitting procedure. The individual vibrational constants are listed in Table 5.

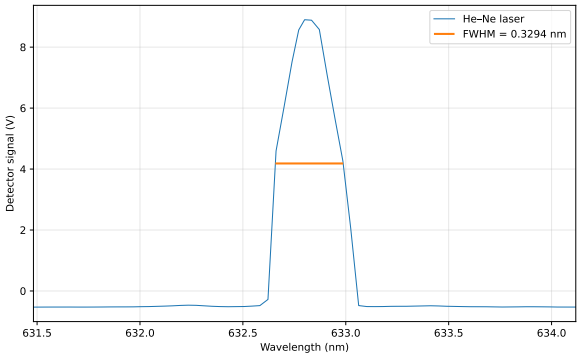
For the $v'' = 0$ progression, the harmonic constants from WIn4902 and WIn6403 differ from WIn4903 by approximately -0.38% and $+1.31\%$, respectively. For $v'' = 1$, the corresponding differences are approximately $+1.8\%$ and $+1.6\%$. These percentages describe the harmonic constants; the other derived quantities have different scan-to-scan variations. These differences support the repeatability of the harmonic-constant estimates.

Table 5. Vibrational constants from the three scans. Both constants are expressed in cm^{-1}.

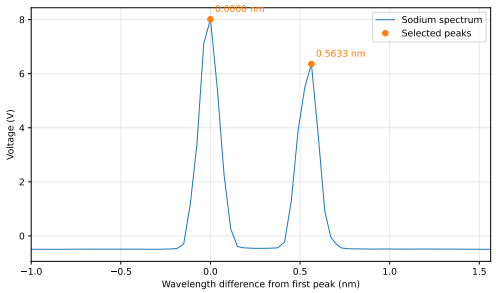
Scan	v''	ω'_e	$\omega'_e x'_e$
WIn4902	0	126.6 ± 1.1	0.9946 ± 0.0094
WIn4903	0	127.1 ± 1.1	1.0060 ± 0.0089
WIn6403	0	128.72 ± 0.84	1.0159 ± 0.0071
WIn4902	1	121.1 ± 2.3	0.810 ± 0.065
WIn4903	1	118.9 ± 1.3	0.750 ± 0.033
WIn6403	1	120.8 ± 1.2	0.797 ± 0.028

B Resolution measurements

The He–Ne and sodium measurements used to assess spectral resolution are shown in **(Fig. 6)**. The laser linewidth was also used to assign the shared reference-position uncertainty.



(a)



(b)

Fig. 6. Resolution measurements. (a) He–Ne laser profile. (b) Sodium doublet, with wavelength difference measured from the first maximum. Orange markers identify the selected sodium peaks.

C Data acquisition

In [23]:

```
from pydaqmx_helper.adc import ADC
import numpy as np
import matplotlib.pyplot as plt
import time


# Set up ADC
myADC = ADC()
myADC.addChannels(
    [0],
    ADC_mode="DAQmx_Val_Diff",
    minRange=-10.0,
    maxRange=10.0
)

# Acquisition settings
n_samples = 20
sample_rate = 10000       # Hz = 0.002 seconds
sleep_time = 0.001 # seconds = 0.003 seconds per cycle #~ x 1.8 nm/ s = 0.0055 n
duration = 105

times = []
voltages = []
print("Ready to record")
input("Press Enter to Start")
print("Recording...")

t0 = time.perf_counter()

while time.perf_counter() - t0 < duration:

    # Time immediately before sampling
    t_before = time.perf_counter()

    # Acquire 50 samples at 1000 Hz
    data = myADC.sampleVoltages(n_samples, sample_rate)
    samples = np.array(data[0], dtype=float)

    # Time immediately after sampling
    t_after = time.perf_counter()

    # Average the 50 voltage measurements
    mean_voltage = samples.mean()
    standard_deviation=samples.std()

    # Timestamp the middle of the acquisition block
    t_measurement = ((t_before + t_after) / 2) - t0

    times.append(t_measurement)
    voltages.append(mean_voltage)

    # Wait before taking the next block
    time.sleep(sleep_time)

print("Finished.")
np.savetxt(
    "mercury6208.csv",
    np.column_stack((times,voltages)),
```

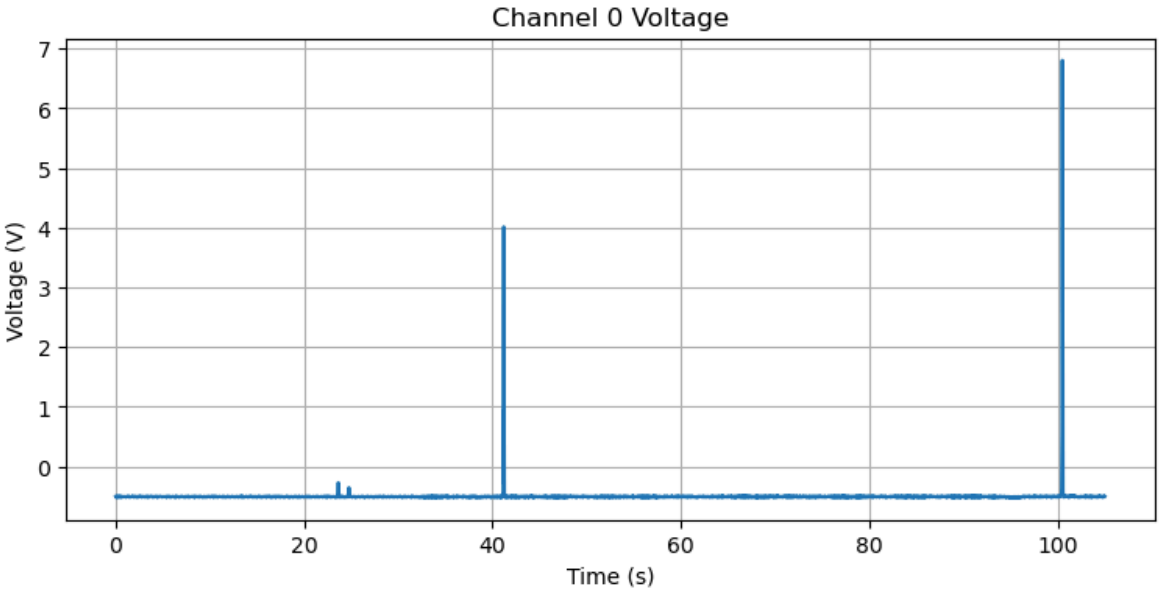
```
    delimiter=",",
    header="time_s,voltage_V",
    comments=""
)

times = np.array(times)
voltages = np.array(voltages)

print(f"Number of recorded points = {len(voltages)}")
print(f"Mean voltage = {voltages.mean():.6f} V")

# Plot
plt.figure(figsize=(9, 4))
plt.plot(times, voltages)
plt.xlabel("Time (s)")
plt.ylabel("Voltage (V)")
plt.title("Channel 0 Voltage")
plt.grid(True)
plt.show()
#06V,0.54+-0.01A
```

```
Activated Channel 0
Ready to record
Recording...
Finished.
Number of recorded points = 5400
Mean voltage = -0.498175 V
```



D Data Analysis

The Jupyter notebook used for calibration, peak selection, uncertainty propagation and Birge–Sponer analysis is reproduced below.

In [4]:
```
import csv, json, itertools, hashlib
import numpy as np
from scipy.signal import find_peaks, savgol_filter, medfilt
import matplotlib.pyplot as plt
from pathlib import Path

folder = Path(
    r"C:\Users\lewis\OneDrive\Documents\Lab reports\Iodine Spectrometry")
```

In [2]:
```
# filenames wrong


hgn490true =["mercury4304.csv",
             "mercury4305.csv",
             "mercury4306.csv",
             "mercury4307.csv",
               ]
hgn640true =["mercury6205.csv",
             "mercury6206.csv",
             "mercury6207.csv",
             "mercury6208.csv",
             ]
hgntruecombined =hgn490true + hgn640true
```

Mercury Calibration

In [7]:
```
for filename in hgn490true:
    data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
    plt.plot(data[:, 0], data[:, 1], label=filename)

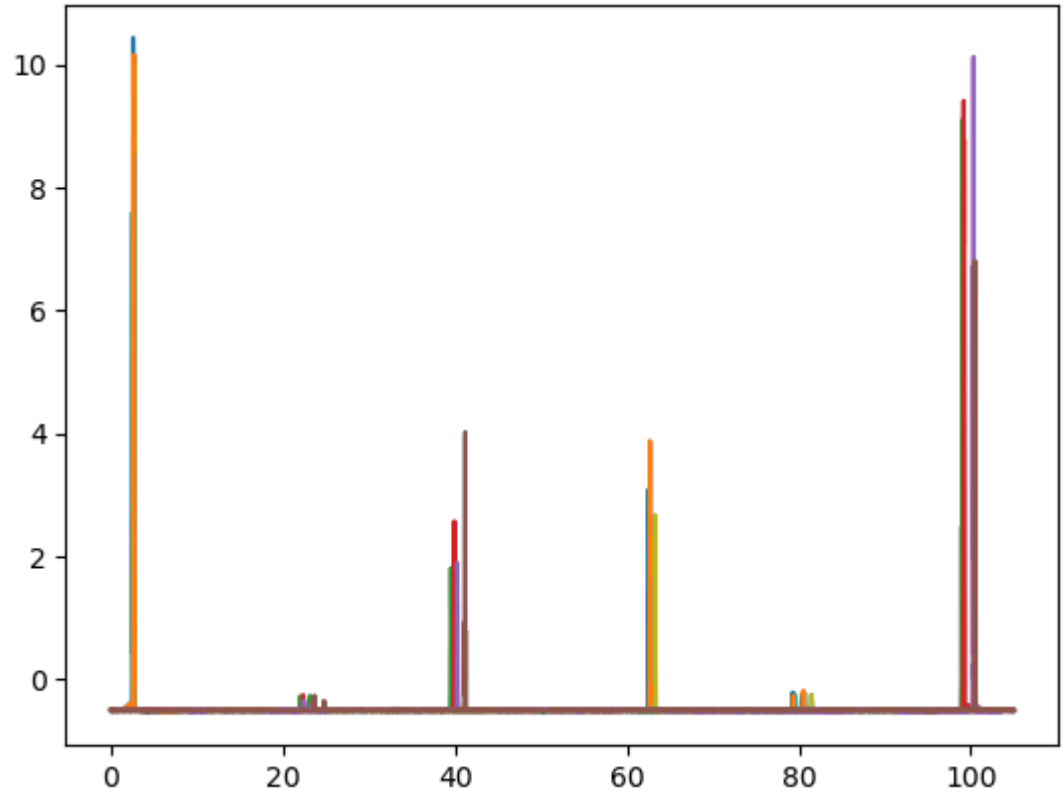
for filename in hgn640true:
    data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
    plt.plot(data[:, 0], data[:, 1], label=filename)

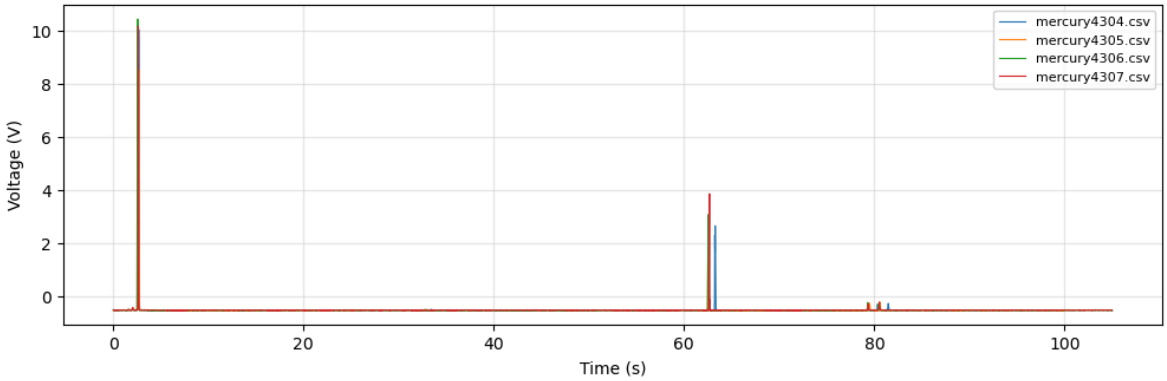
for filename in hgntruecombined:
    data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
    plt.plot(data[:, 0], data[:, 1], label=filename)

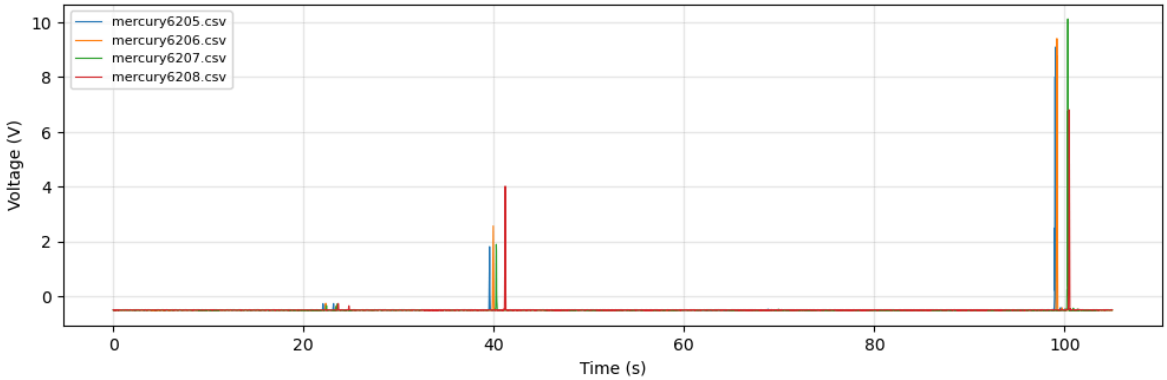
fig, axes = plt.subplots(3, 1, figsize=(12, 12))

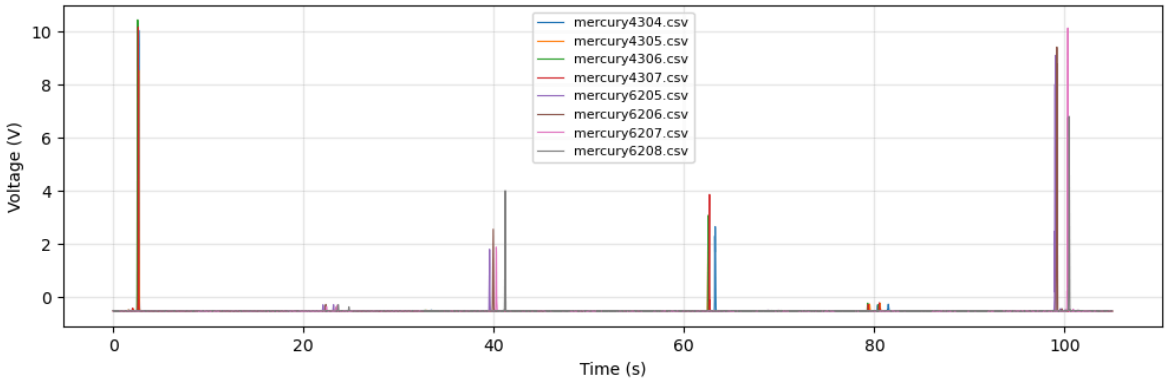
groups = [
    (hgn490true, "Forward scans"),
    (hgn640true, "Reverse scans"),
    (hgntruecombined, "All scans"),
]

for ax, (files, title) in zip(axes, groups):
    for filename in files:
        data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
        ax.plot(data[:, 0], data[:, 1], label=filename, linewidth=0.8)
    ax.set_xlabel("Time (s)")
    ax.set_ylabel("Voltage (V)")
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3)
```


0
2
4
6
8
10
0
20
40
60
80
100


mercury4304.csv
mercury4305.csv
mercury4306.csv
mercury4307.csv
Voltage (V)
Time (s)


mercury6205.csv
mercury6206.csv
mercury6207.csv
mercury6208.csv
Voltage (V)
Time (s)


mercury4304.csv
mercury4305.csv
mercury4306.csv
mercury4307.csv
mercury6205.csv
mercury6206.csv
mercury6207.csv
mercury6208.csv
Voltage (V)
Time (s)

In [337…

```
from scipy.optimize import curve_fit

def straight(x, intercept, slope):
    return intercept + slope*x

# Detected peak_number -> mercury wavelength (nm)
B, G, Y1, Y2 = 435.835, 546.0735, 576.9598, 579.0663

line_assignments = {
    "mercury4304.csv": {0: B, 1: G, 2: Y1, 3: Y2},
    "mercury4305.csv": {0: B, 1: G, 2: Y1, 3: Y2},
    "mercury4306.csv": {0: B, 1: G, 2: Y1, 3: Y2},
    "mercury4307.csv": {0: B, 1: G, 2: Y1, 3: Y2},

    "mercury6205.csv": {0: Y2, 1: Y1, 2: G, 3: B},
    "mercury6206.csv": {0: Y2, 1: Y1, 2: G, 3: B},
    "mercury6207.csv": {0: Y2, 1: Y1, 2: G, 4: B},
    "mercury6208.csv": {0: Y2, 1: Y1, 2: G, 3: B},
}

results = []
calibration_covariances = {}
calibration_peak_tables = []

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

for ax, files, direction in zip(
    axes,
    [hgn490true, hgn640true],
    ["Forward", "Reverse"]
):
    scans = []

    for filename in files:
        assignment = line_assignments[filename]

        peaks = peak_table[
            (peak_table["file"] == filename)
            & peak_table["peak_number"].isin(assignment)
        ].copy()

        peaks = peaks.sort_values("peak_time_s")
        peaks["wavelength_nm"] = peaks["peak_number"].map(assignment)
        scans.append(peaks)

    scans = pd.concat(scans, ignore_index=True)

    mean_width = scans.groupby("wavelength_nm")["FWHM_s"].mean()
    scans["u_time_s"] = scans["wavelength_nm"].map(mean_width) / 2
    calibration_peak_tables.append(scans)

    print(direction, "— mean FWHM / 2 in seconds:")
    display(mean_width / 2)

    for filename in files:
        peaks = scans[scans["file"] == filename]

        wavelength = peaks["wavelength_nm"].to_numpy()
        time = peaks["peak_time_s"].to_numpy()
```

```
        u_time = peaks["u_time_s"].to_numpy()

        if not np.all(np.isfinite(u_time) & (u_time > 0)):
            raise ValueError(f"Check the measured widths for {filename}")

        # Fit time = alpha + beta*wavelength.
        parameters, covariance = curve_fit(
            straight,
            wavelength,
            time,
            sigma=u_time,
            absolute_sigma=True
        )

        alpha, beta = parameters

        # Convert to wavelength = a + b*time.
        a = -alpha / beta
        b = 1 / beta

        J = np.array([
            [-1 / beta, alpha / beta**2],
            [0,         -1 / beta**2]
        ])

        covariance_ab = J @ covariance @ J.T
        ua, ub = np.sqrt(np.diag(covariance_ab))

        calibration_covariances[filename] = covariance_ab
        results.append([filename, a, ua, b, ub])

        grid = np.linspace(time.min(), time.max(), 100)
        fitted_line, = ax.plot(
            grid, a + b*grid, label=filename
        )

        ax.errorbar(
            time,
            wavelength,
            xerr=u_time,
            fmt="o",
            color=fitted_line.get_color(),
            markersize=4,
            capsize=5,
            capthick=1.5,
            elinewidth=1.5
        )

    ax.set_title(direction)
    ax.set_xlabel("Time (s)")
    ax.set_ylabel("Wavelength (nm)")
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3)
plt.savefig('calibration.pdf')
plt.tight_layout()
plt.show()

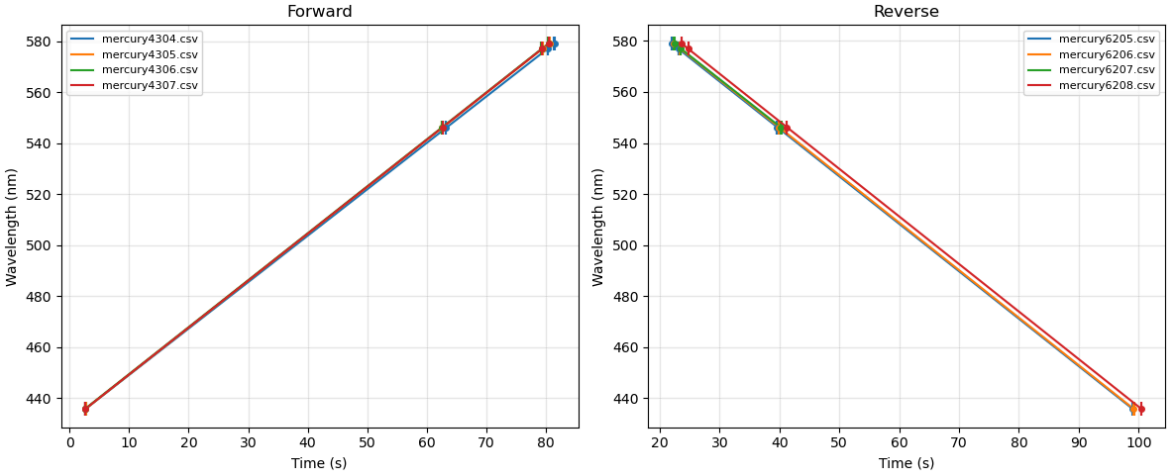
calibration = pd.DataFrame(
    results,
    columns=["file", "a_nm", "u_a_nm", "b_nm_s", "u_b_nm_s"]
```

```
)

# Correctly assigned peaks retained for subsequent residual plots.
calibration_peaks = pd.concat(
    calibration_peak_tables, ignore_index=True
)

display(calibration.round(5))
```

```
Forward — mean FWHM / 2 in seconds:
wavelength_nm
435.8350    0.027553
546.0735    0.025117
576.9598    0.031286
579.0663    0.030512
Name: FWHM_s, dtype: float64
Reverse — mean FWHM / 2 in seconds:
wavelength_nm
435.8350    0.028516
546.0735    0.028557
576.9598    0.029902
579.0663    0.038067
Name: FWHM_s, dtype: float64
```



	file	a_nm	u_a_nm	b_nm_s	u_b_nm_s
0	mercury4304.csv	430.99080	0.05090	1.81762	0.00080
1	mercury4305.csv	430.96378	0.05147	1.83773	0.00082
2	mercury4306.csv	431.09181	0.05145	1.83873	0.00082
3	mercury4307.csv	430.96833	0.05149	1.83848	0.00082
4	mercury6205.csv	619.88004	0.05221	-1.85911	0.00088
5	mercury6206.csv	620.56485	0.05258	-1.86266	0.00089
6	mercury6207.csv	620.40930	0.12398	-1.84648	0.00397
7	mercury6208.csv	623.04027	0.05360	-1.86365	0.00089

In [354…

```
import numpy as np
import pandas as pd

cal = calibration.copy()
```

```
cal["direction"] = np.where(
    cal["b_nm_s"] > 0, "Forward", "Reverse"
)

speed_summary = cal.groupby("direction").agg(
    N=("b_nm_s", "count"),
    b=("b_nm_s", "mean"),
    SD=("b_nm_s", "std")
)

# Uncertainty in the mean calibration speed.
speed_summary["u_mean"] = (
    speed_summary["SD"] / np.sqrt(speed_summary["N"])
)

# Uncertainty when applying that speed to a separate iodine scan.
speed_summary["u_b"] = np.sqrt(
    speed_summary["SD"]**2 + speed_summary["u_mean"]**2
)

display(speed_summary.round(6))

for direction, row in speed_summary.iterrows():
    print(f"{direction}: b = {row['b']:.5f} ± {row['u_b']:.5f} nm/s")
```

direction	N	b	SD	u_mean	u_b
Forward	4	1.833137	0.010355	0.005178	0.011577
Reverse	4	-1.857976	0.007905	0.003953	0.008838

```
Forward: b = 1.83314 ± 0.01158 nm/s
Reverse: b = -1.85798 ± 0.00884 nm/s
```

In [347…

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

for ax, files, direction in zip(
    axes,
    [hgn490true, hgn640true],
    ["Forward", "Reverse"]
):
    for filename in files:
        peaks = calibration_peaks[
            calibration_peaks["file"] == filename
        ].sort_values("peak_time_s")

        time = peaks["peak_time_s"].to_numpy()
        wavelength = peaks["wavelength_nm"].to_numpy()
        u_time = peaks["u_time_s"].to_numpy()

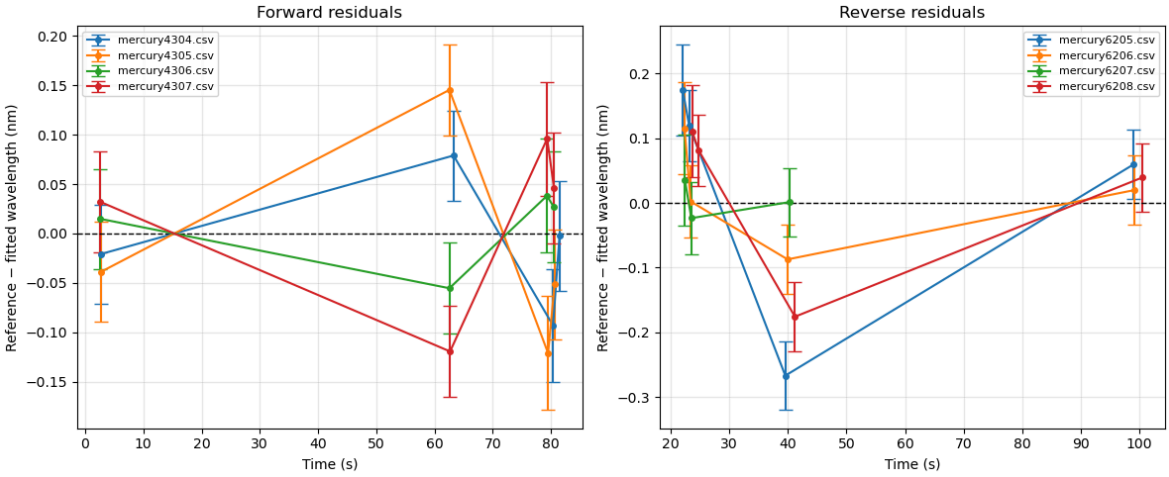
        fit = calibration.loc[calibration["file"] == filename].iloc[0]
        a = fit["a_nm"]
        b = fit["b_nm_s"]

        residual = wavelength - (a + b*time)
        u_position = abs(b) * u_time

        ax.errorbar(
```

```
            time,
            residual,
            yerr=u_position,
            fmt="o-",
            markersize=4,
            capsize=5,
            label=filename
        )

    ax.axhline(0, color="black", linestyle="--", linewidth=1)
    ax.set_title(direction + " residuals")
    ax.set_xlabel("Time (s)")
    ax.set_ylabel("Reference − fitted wavelength (nm)")
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3)
plt.savefig('residuals.pdf')
plt.tight_layout()
plt.show()
```



Sodium and Ne He Resolution

In [348…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import find_peaks

filename = "na4304.csv"

# Mean forward speed from mercury calibration.
b = calibration.loc[
    calibration["file"].isin(hgn490true), "b_nm_s"
].mean()

data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
data = data[np.argsort(data[:, 0])]
time, voltage = data[:, 0], data[:, 1]

peaks, properties = find_peaks(
    voltage, prominence=0.1, distance=3
)

chosen = peaks[np.argsort(properties["prominences"])[-2:]]
chosen = chosen[np.argsort(time[chosen])]
```

```
# Set the first sodium peak to zero wavelength difference.
delta_wavelength = b * (time - time[chosen[0]])
separation = delta_wavelength[chosen[1]]

fig, ax = plt.subplots(figsize=(9, 5))

ax.plot(
    delta_wavelength, voltage,
    color="tab:blue",
    linewidth=0.8,
    label="Sodium spectrum"
)

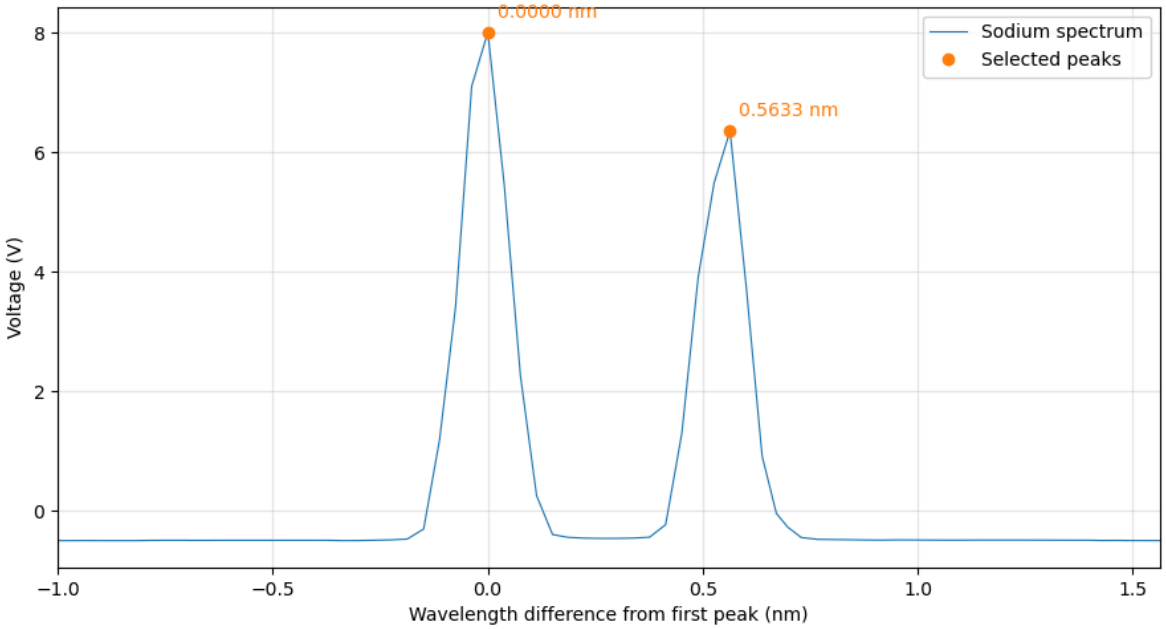
ax.plot(
    delta_wavelength[chosen], voltage[chosen],
    linestyle="none",
    marker="o",
    color="tab:orange",
    markersize=6,
    label="Selected peaks"
)

for k in chosen:
    ax.annotate(
        f"{delta_wavelength[k]:.4f} nm",
        (delta_wavelength[k], voltage[k]),
        xytext=(5, 8),
        textcoords="offset points",
        color="tab:orange"
    )

ax.set_xlim(-1, separation + 1)
ax.set_xlabel("Wavelength difference from first peak (nm)")
ax.set_ylabel("Voltage (V)")
ax.grid(alpha=0.3)
ax.legend()
plt.savefig('sodium.pdf')
plt.tight_layout()
plt.show()

sodium_results = pd.DataFrame([{
    "file": filename,
    "Hg_speed_nm_s": b,
    "separation_nm": separation,
    "reference_separation_nm": 0.5974,
    "separation_difference_nm": separation - 0.5974
}])

display(sodium_results.round(5))
```



	file	Hg_speed_nm_s	separation_nm	reference_separation_nm	separation_differe
0	na4304.csv	1.83314	0.56327	0.5974	

In [336…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import find_peaks, peak_widths
from pathlib import Path

filename = "nehe640.csv"
b, u_b = calibration_speeds["Reverse"]

data = np.loadtxt(folder / filename, delimiter=",", skiprows=1)
data = data[np.argsort(data[:, 0])]
time, signal = data[:, 0], data[:, 1]

peaks, properties = find_peaks(signal, prominence=0)
if len(peaks) == 0:
    raise ValueError(f"No positive laser peak found in {filename}")

peak = peaks[np.argmax(properties["prominences"])]

_, heights, left_ips, right_ips = peak_widths(
    signal, [peak], rel_height=0.5
)

samples = np.arange(len(time))
left_time = np.interp(left_ips[0], samples, time)
right_time = np.interp(right_ips[0], samples, time)

fwhm_s = right_time - left_time
fwhm_nm = abs(b) * fwhm_s
u_laser_position_nm = fwhm_nm / 2

# Place the selected laser maximum at 632.8 nm.
wavelength = 632.8 + b * (time - time[peak])
width_limits_nm = np.sort(
```

```
    632.8 + b * (np.array([left_time, right_time]) - time[peak])
)
order = np.argsort(wavelength)

fig, ax = plt.subplots(figsize=(8, 5))

ax.plot(
    wavelength[order], signal[order],
    color="tab:blue", linewidth=1,
    label="He–Ne laser"
)
ax.hlines(
    heights[0],
    width_limits_nm[0],
    width_limits_nm[1],
    color="tab:orange",
    linewidth=2,
    label=f"FWHM = {fwhm_nm:.4f} nm"
)

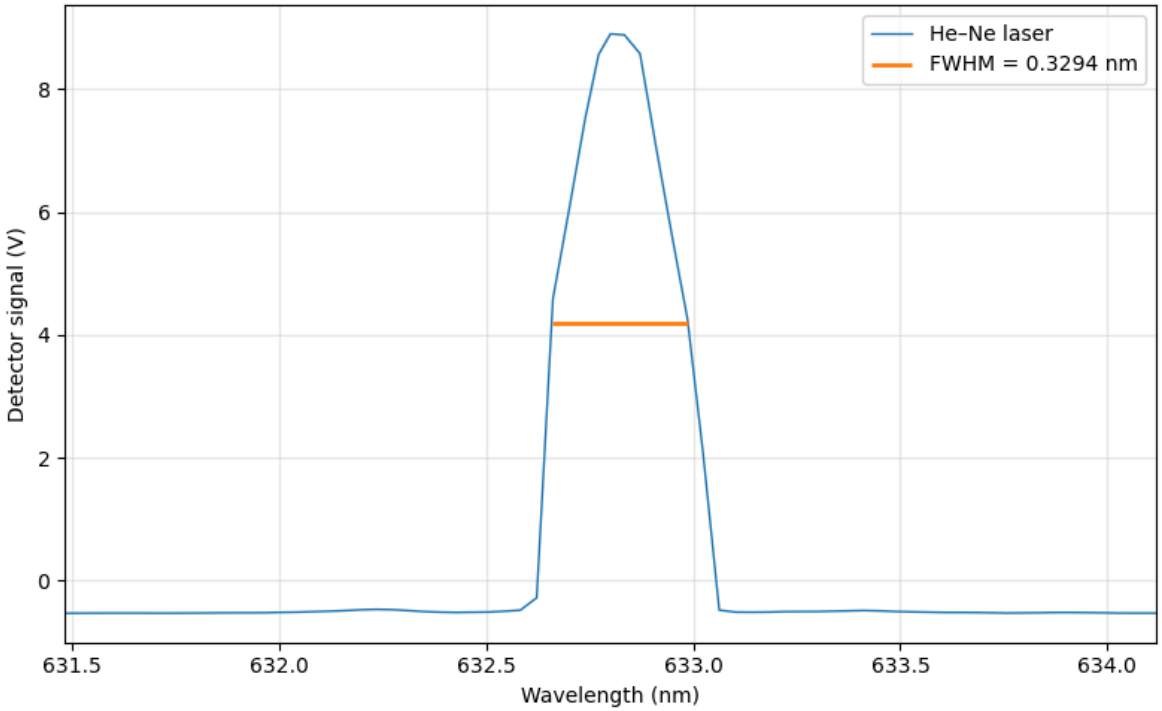
half_window = max(
    4 * fwhm_nm,
    10 * abs(b) * np.median(np.diff(time))
)
ax.set_xlim(632.8 - half_window, 632.8 + half_window)
ax.set_xlabel("Wavelength (nm)")
ax.set_ylabel("Detector signal (V)")
ax.ticklabel_format(axis="x", style="plain", useOffset=False)
ax.grid(alpha=0.3)
ax.legend()

fig.tight_layout()


fig.savefig('nehe.pdf')

plt.show()

display(pd.DataFrame([{
    "file": filename,
    "FWHM_s": fwhm_s,
    "FWHM_nm": fwhm_nm,
    "u_laser_position_nm": u_laser_position_nm,
}]).round(6))
```



	file	FWHM_s	FWHM_nm	u_laser_position_nm
0	nehe640.csv	0.176037	0.329423	0.164712

Absorbance Spectrum

In [361…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import medfilt

pairs = [
    ("WIn4902.csv", "Wn4902.csv", "Forward"),
    ("WIn4903.csv", "Wn4903.csv", "Forward"),
    ("WIn6403.csv", "Wn6403.csv", "Reverse"),
]


scan_speeds = {
    "Forward": calibration_speeds["Forward"][0],
    "Reverse": calibration_speeds["Reverse"][0],
}

dark_voltage = 0.0


def calibrated_spectrum(filename, direction):
    time, voltage = np.loadtxt(
        folder / filename, delimiter=",", skiprows=1
    ).T

    b = scan_speeds[direction]

    laser_signal = voltage - medfilt(voltage, 31)
```

```
    if direction == "Forward":
        search = np.flatnonzero(time > 60)
    else:
        search = np.flatnonzero(time < 12)

    if len(search) == 0:
        raise ValueError(f"No laser search interval in {filename}")

    k = search[np.argmax(laser_signal[search])]

    if laser_signal[k] < 0.1:
        raise ValueError(f"No clear laser marker in {filename}")

    background = (
        (np.abs(time - time[k]) > 0.4)
        & (np.abs(time - time[k]) < 0.8)
    )
    baseline = np.median(voltage[background])

    nearby = np.abs(time - time[k]) < 0.22
    signal = np.maximum(voltage[nearby] - baseline, 0)

    area = np.trapezoid(signal, time[nearby])
    if not np.isfinite(area) or area <= 0:
        raise ValueError(f"Cannot calculate laser centre in {filename}")

    laser_time = (
        np.trapezoid(time[nearby] * signal, time[nearby]) / area
    )

    wavelength = 632.8 + b * (time - laser_time)
    order = np.argsort(wavelength)

    return wavelength[order], voltage[order]


absorption_results = {}

fig, axes = plt.subplots(2, 1, figsize=(12, 9), sharex=True)

for iodine_file, tungsten_file, direction in pairs:
    wi, vi = calibrated_spectrum(iodine_file, direction)
    ww, vw = calibrated_spectrum(tungsten_file, direction)

    low = max(500, wi.min(), ww.min())
    high = min(625, wi.max(), ww.max())

    use = (wi >= low) & (wi <= high)
    wavelength = wi[use]
    iodine_voltage = vi[use]

    tungsten_voltage = np.interp(wavelength, ww, vw)
    voltage_difference = tungsten_voltage - iodine_voltage

    iodine_corrected = iodine_voltage - dark_voltage
    tungsten_corrected = tungsten_voltage - dark_voltage

    absorbance = np.full(len(wavelength), np.nan)
    valid = (iodine_corrected > 0) & (tungsten_corrected > 0)
```

```
    absorbance[valid] = -np.log10(
        iodine_corrected[valid] / tungsten_corrected[valid]
    )

    name = iodine_file.replace(".csv", "")

    absorption_results[name] = pd.DataFrame({
        "wavelength_nm": wavelength,
        "iodine_voltage_V": iodine_voltage,
        "tungsten_voltage_V": tungsten_voltage,
        "voltage_difference_V": voltage_difference,
        "absorbance": absorbance,
    })

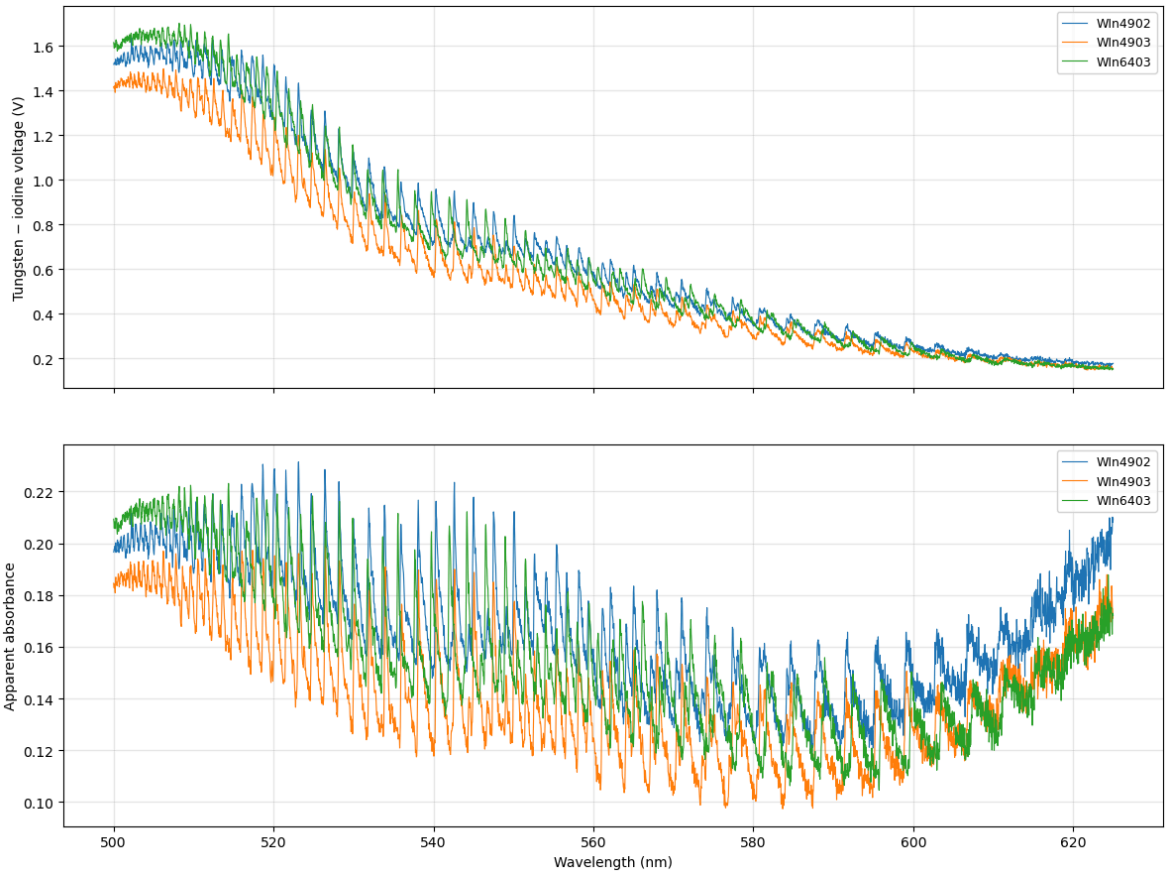
    axes[0].plot(
        wavelength, voltage_difference,
        linewidth=0.8, label=name
    )
    axes[1].plot(
        wavelength, absorbance,
        linewidth=0.8, label=name
    )

    absorption_results[name].to_csv(
        name + "_absorption.csv", index=False
    )

axes[0].set_ylabel("Tungsten − iodine voltage (V)")
axes[1].set_ylabel("Apparent absorbance")
axes[1].set_xlabel("Wavelength (nm)")

for ax in axes:
    ax.legend(fontsize=9)
    ax.grid(alpha=0.3)

fig.tight_layout()
fig.subplots_adjust(hspace=0.15)
fig.savefig("absorbance.pdf", bbox_inches="tight")
plt.show()
```



Peak assignment

In [327…

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import find_peaks, savgol_filter

groups4903 = {
    "v'' = 0": [
        19,21,23,25,27,29,31,33,35,37,39,40,41,42,43,44,
        45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,
        61,62,63,64
    ],
    "v'' = 1": [
        5,6,7,8,9,10,11,12,13,15,16,17,18,20,22,24,26,28,30,32,34,36,38
    ],
}

b, u_b = calibration_speeds["Forward"]
laser_nm = 632.8

data = absorption_results["WIn4903"].sort_values("wavelength_nm")
wavelength = data["wavelength_nm"].to_numpy()
absorbance = data["absorbance"].to_numpy()
wavenumber = 1e7 / wavelength

if not np.all(np.isfinite(absorbance)):
    raise ValueError("Check invalid absorbance values in WIn4903")


smooth = savgol_filter(absorbance, 7, 2)
distance = max(
```

```
    1, int(np.ceil(0.4 / np.median(np.diff(wavelength))))
)

peaks, _ = find_peaks(
    smooth, prominence=0.01, distance=distance
)
peaks = peaks[
    (wavenumber[peaks] >= 16000)
    & (wavenumber[peaks] <= 20000)
]
peaks = peaks[np.argsort(wavenumber[peaks])]


u_wavelength = np.abs((wavelength - laser_nm) / b) * u_b
u_wavenumber = 1e7 * u_wavelength / wavelength**2

fig, ax = plt.subplots(figsize=(16, 5))

ax.plot(
    wavenumber / 1e4,
    absorbance,
    color="0.2",
    linewidth=0.7,
    label="Absorbance"
)

styles = {
    "v'' = 0": ("tab:blue", "x"),
    "v'' = 1": ("tab:orange", "o"),
}

for label, numbers in groups4903.items():
    colour, marker = styles[label]
    numbers = np.asarray(numbers, dtype=int)

    if np.any((numbers < 0) | (numbers >= len(peaks))):
        raise ValueError(f"Check selected peak numbers for {label}")

    selected = peaks[numbers]

    ax.errorbar(
        wavenumber[selected] / 1e4,
        smooth[selected],
        xerr=u_wavenumber[selected] / 1e4,
        fmt=marker,
        linestyle="none",
        color=colour,
        markersize=5,
        capsize=3,
        elinewidth=0.8,
        label=label
    )


    for number, k in zip(numbers, selected):
        ax.annotate(
            str(number),
            (wavenumber[k] / 1e4, smooth[k]),
            xytext=(0, 6),
            textcoords="offset points",
```

```
            ha="center",
            rotation=90,
            fontsize=7,
            color=colour
        )


reference_bands = [
    (0, 23, 553.400),
    (0, 18,  567.82,),
    (0, 27,  543.47,),
    (0, 28,  541.18,),
    (0, 29,   539.89,),
    (0, 30, 536.866),
    (1, 15, 581.278),
    (1, 18, 571.60),
    (1, 19, 568.60),
    (1, 24, 554.758),
]


wavelength_convention = "air"
n_air = 1.0002680  # Approximate refractive index used in the source.

labelled = set()

for v_lower, v_upper, lambda_vac in reference_bands:
    colour = "tab:blue" if v_lower == 0 else "tab:orange"

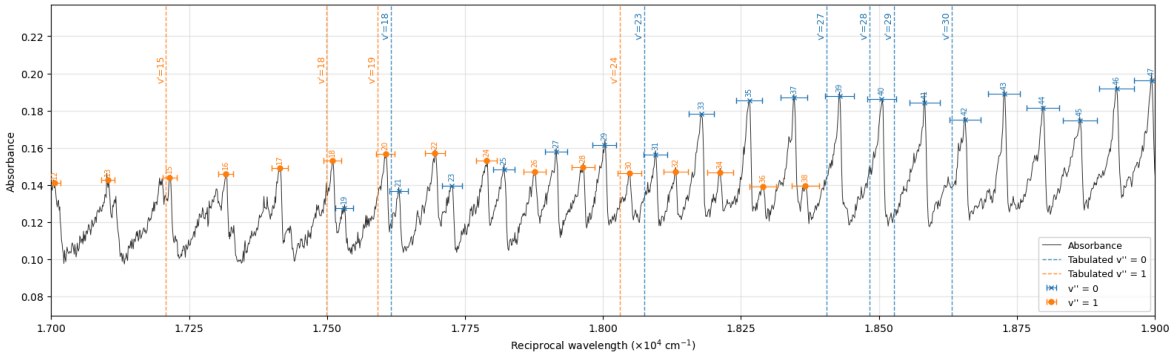
    lambda_plot = (
        lambda_vac / n_air
        if wavelength_convention == "air"
        else lambda_vac
    )
    x_reference = (1e7 / lambda_plot) / 1e4

    ax.axvline(
        x_reference,
        color=colour,
        linestyle="--",
        linewidth=1,
        alpha=0.8,
        label=(
            f"Tabulated v'' = {v_lower}"
            if v_lower not in labelled
            else "_nolegend_"
        )
    )

    ax.text(
        x_reference,
        0.98 if v_lower == 0 else 0.84,
        f"v'={v_upper}",
        transform=ax.get_xaxis_transform(),
        rotation=90,
        ha="right",
        va="top",
        fontsize=9,
        color=colour
    )
```

```
    labelled.add(v_lower)

ax.set_xlabel(
    r"Reciprocal wavelength ($\times 10^4$ cm$^{-1}$)"
    if wavelength_convention == "air"
    else r"Wavenumber ($\times 10^4$ cm$^{-1}$)"
)
ax.set_ylabel("Absorbance")
ax.set_xlim(1.7, 1.9)
ax.margins(y=0.25)
ax.grid(alpha=0.3)
ax.legend(fontsize=9)
plt.savefig('methods.pdf')
plt.tight_layout()
plt.show()
```



Birge Sponer

In [366…

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import find_peaks, savgol_filter

blue_anchor_v = 30
orange_anchor_nm = 581.278
orange_anchor_v = 15

# He–Ne position uncertainty: half the measured FWHM.
laser_nm = 632.8
laser_fwhm_nm = 0.3294
u_laser_nm = laser_fwhm_nm / 2  # 0.1647 nm

# Exclude this orange spacing from all three fits.
excluded_sequence = "v'' = 1"
excluded_midpoint = 14.5

directions = {
    "WIn4902": "Forward",
    "WIn4903": "Forward",
    "WIn6403": "Reverse",
}

colours = {
    "v'' = 0": "tab:blue",
    "v'' = 1": "tab:orange",
```

```
    "v'' = 2": "tab:green",
}

fit_results = []
spacing_tables = {}
fit_covariances = {}

for name, selected_groups in all_groups.items():

    b, u_b = calibration_speeds[directions[name]]

    data = absorption_results[name].sort_values("wavelength_nm")
    wavelength = data["wavelength_nm"].to_numpy()
    absorbance = data["absorbance"].to_numpy()
    wavenumber = 1e7 / wavelength

    smooth = savgol_filter(absorbance, 7, 2)
    distance = max(
        1, int(np.ceil(0.4 / np.median(np.diff(wavelength))))
    )

    peaks, _ = find_peaks(
        smooth, prominence=0.01, distance=distance
    )
    peaks = peaks[
        (wavenumber[peaks] >= 16000)
        & (wavenumber[peaks] <= 20000)
    ]
    peaks = peaks[np.argsort(wavenumber[peaks])]

    fig, ax = plt.subplots(figsize=(11, 6))

    for label, numbers in selected_groups.items():

        colour = colours[label]
        selected = peaks[np.asarray(numbers, dtype=int)]
        selected = selected[np.argsort(wavenumber[selected])]

        nu = wavenumber[selected]
        lam = wavelength[selected]

        if len(nu) < 4:
            raise ValueError(
                f"{name}, {label}: at least four bands are needed."
            )

        n = np.arange(len(nu), dtype=float)

        if label == "v'' = 0":
            anchor_n = np.flatnonzero(
                selected == peaks[anchor_peak[name]]
            ).item()

            v = n + (blue_anchor_v - anchor_n)
            numbering = "absolute"

            print(
                f"{name}: blue peak {anchor_peak[name]} "
                f"at {lam[anchor_n]:.4f} nm "
                f"assigned v'={blue_anchor_v}"
```

```
        )

    elif label == "v'' = 1":
        anchor_n = np.argmin(np.abs(lam - orange_anchor_nm))
        orange_peak_number = np.flatnonzero(
            peaks == selected[anchor_n]
        ).item()

        v = n + (orange_anchor_v - anchor_n)
        numbering = "absolute"

        print(
            f"{name}: orange peak {orange_peak_number} "
            f"at {lam[anchor_n]:.4f} nm "
            f"(nearest {orange_anchor_nm} nm) "
            f"assigned v'={orange_anchor_v}"
        )

    else:
        v = n
        numbering = "relative"

    midpoint = (v[:-1] + v[1:]) / 2
    delta_G = np.diff(nu)

    # Preserve existing independent peak-scatter estimate.
    quadratic = np.polyfit(n, nu, 2)
    residual = nu - np.polyval(quadratic, n)
    scatter = np.sqrt(
        np.sum(residual**2) / (len(nu) - 3)
    )

    sample_steps = np.abs(np.gradient(wavenumber))[selected]
    u_sampling = sample_steps / np.sqrt(12)
    u_peak = np.maximum(scatter, u_sampling)

    # Sensitivity to the shared mercury scan speed.
    dnu_db = -1e7 * (lam - laser_nm) / (b * lam**2)

    # Sensitivity to a shared additive wavelength-reference offset.
    dnu_dlaser = -1e7 / lam**2

    # Band-position covariance:
    # independent peak scatter + shared speed + shared laser position.
    # The laser term is included once, as FWHM/2 in wavelength units.
    C_nu = (
        np.diag(u_peak**2)
        + u_b**2 * np.outer(dnu_db, dnu_db)
        + u_laser_nm**2 * np.outer(dnu_dlaser, dnu_dlaser)
    )

    # Propagate to adjacent spacings, retaining correlations.
    D = np.diff(np.eye(len(nu)), axis=0)
    C_gap = D @ C_nu @ D.T
    u_delta_G = np.sqrt(np.diag(C_gap))

    keep = np.ones(len(midpoint), dtype=bool)

    if label == excluded_sequence:
        exclude = np.isclose(midpoint, excluded_midpoint)
```

```
        if not np.any(exclude):
            raise ValueError(
                f"No spacing at midpoint {excluded_midpoint} "
                f"in {name}, {label}. Check the exclusion setting."
            )

        keep &= ~exclude
        print(
            f"{name}, {label}: excluded midpoint "
            f"{excluded_midpoint} from the fit."
        )

    # Generalised least-squares fit.
    X = np.column_stack([
        np.ones(len(midpoint)), midpoint
    ])

    X_fit = X[keep]
    y_fit = delta_G[keep]
    C_fit = C_gap[np.ix_(keep, keep)]

    Cinv_X = np.linalg.solve(C_fit, X_fit)
    fit_covariance = np.linalg.inv(X_fit.T @ Cinv_X)

    intercept, slope = (
        fit_covariance
        @ X_fit.T
        @ np.linalg.solve(C_fit, y_fit)
    )

    u_intercept, u_slope = np.sqrt(
        np.diag(fit_covariance)
    )

    # Retain full intercept/slope covariance for later calculations.
    fit_covariances[(name, label)] = fit_covariance.copy()

    ax.errorbar(
        midpoint,
        delta_G,
        yerr=u_delta_G,
        fmt="o",
        color=colour,
        markersize=4,
        capsize=3,
        elinewidth=0.8,
        label=f"{label} ({numbering})"
    )

    if np.any(~keep):
        ax.scatter(
            midpoint[~keep],
            delta_G[~keep],
            marker="x",
            color="black",
            s=70,
            zorder=5,
            label="Excluded from fit"
        )
```

```
        # Fitted line and ±1 standard uncertainty in the fitted mean.
        grid = np.linspace(
            midpoint[keep].min(), midpoint[keep].max(), 200
        )
        X_grid = np.column_stack([
            np.ones(len(grid)), grid
        ])
        fitted_line = intercept + slope * grid

        line_variance = np.sum(
            (X_grid @ fit_covariance) * X_grid, axis=1
        )
        u_line = np.sqrt(np.maximum(line_variance, 0))

        ax.plot(
            grid, fitted_line,
            color=colour,
            linewidth=1.5
        )
        ax.fill_between(
            grid,
            fitted_line - u_line,
            fitted_line + u_line,
            color=colour,
            alpha=0.20
        )

        # intercept = omega_e - omega_exe
        # slope = -2*omega_exe
        if numbering == "absolute":
            omega_e = intercept - slope / 2
            gradient = np.array([1.0, -0.5])
            u_omega_e = np.sqrt(
                gradient @ fit_covariance @ gradient
            )
        else:
            omega_e = np.nan
            u_omega_e = np.nan

        fit_results.append({
            "run": name,
            "sequence": label,
            "numbering": numbering,
            "intercept_cm_inv": intercept,
            "u_intercept": u_intercept,
            "slope_cm_inv": slope,
            "u_slope": u_slope,
            "omega_exe_cm_inv": -slope / 2,
            "u_omega_exe": u_slope / 2,
            "omega_e_cm_inv": omega_e,
            "u_omega_e": u_omega_e,
            "estimated_peak_scatter_cm_inv": scatter,
            "excluded_spacings": int(np.sum(~keep)),
        })

        spacing_tables[(name, label)] = pd.DataFrame({
            "vibrational_midpoint": midpoint,
            "delta_G_cm_inv": delta_G,
            "u_delta_G_cm_inv": u_delta_G,
```

```
            "used_in_fit": keep,
        })

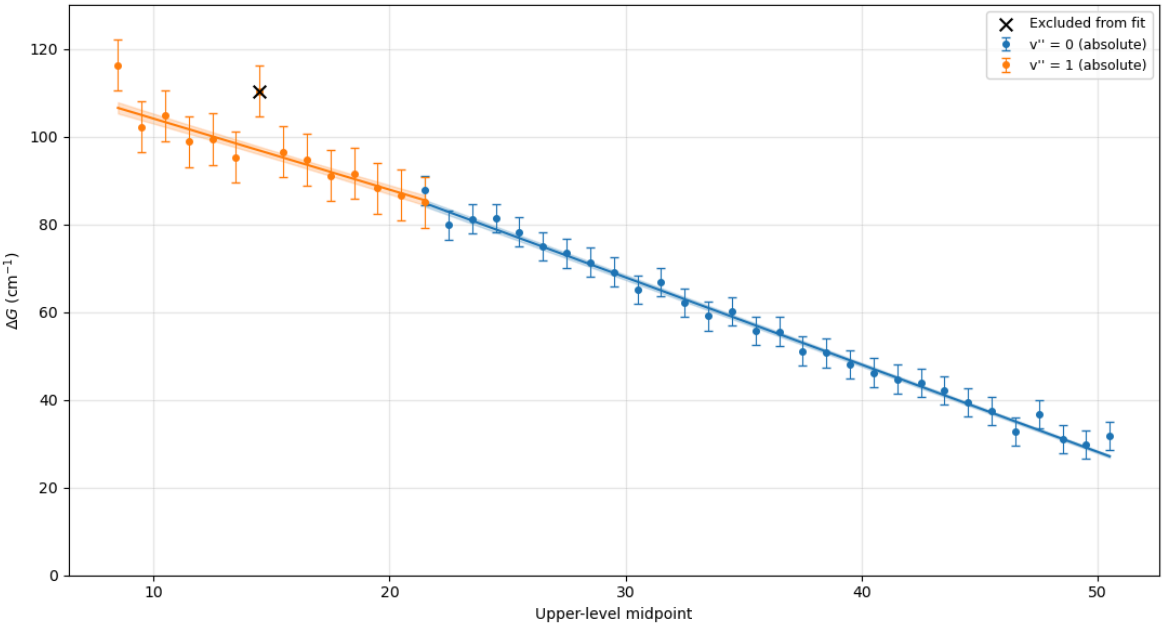
    ax.set_xlabel("Upper-level midpoint")
    ax.set_ylabel(r"$\Delta G$ (cm$^{-1}$)")
    ax.set_ylim(0, 130)
    ax.grid(alpha=0.3)
    ax.legend(fontsize=9)
    fig.tight_layout()

    if name == "WIn4903":
        fig.savefig("WIn4903_birge_sponer.pdf", bbox_inches="tight")
        fig.savefig(
            "WIn4903_birge_sponer.png",
            dpi=300,
            bbox_inches="tight"
        )

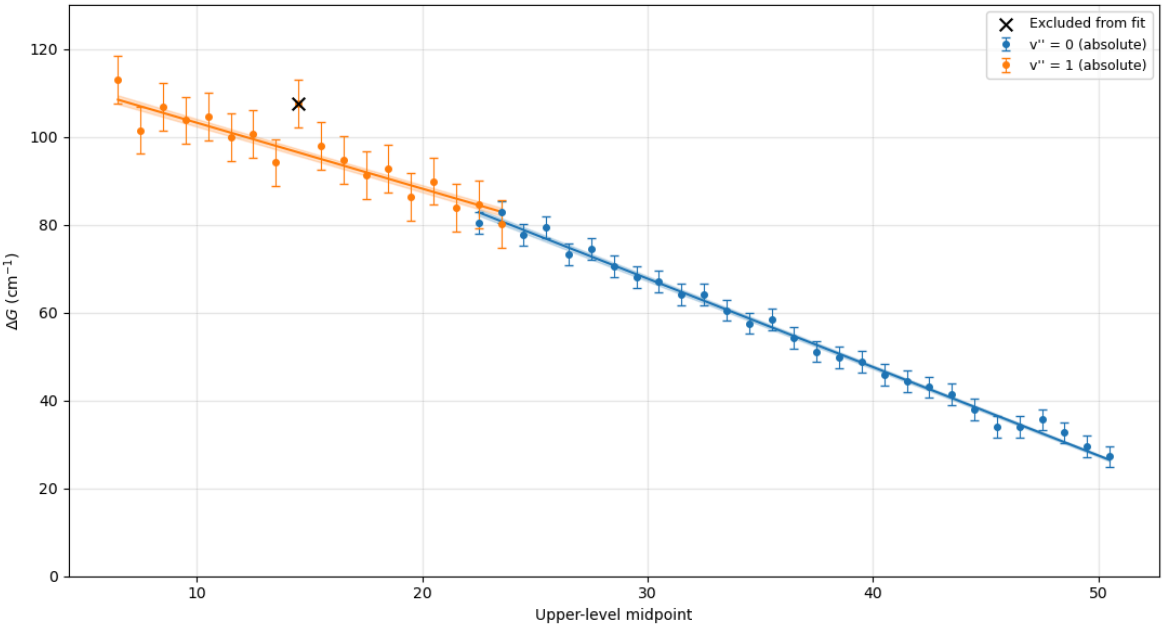
    plt.show()

fit_summary = pd.DataFrame(fit_results)
display(fit_summary.round(5))
```

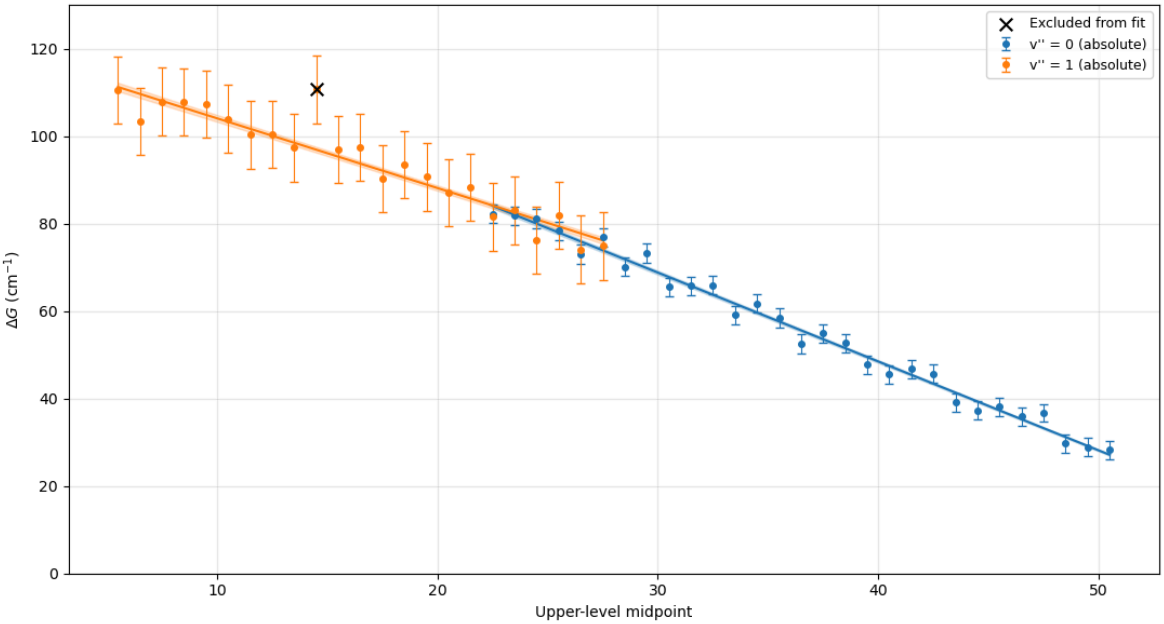
```
WIn4902: blue peak 46 at 536.9262 nm assigned v'=30
WIn4902: orange peak 18 at 582.3828 nm (nearest 581.278 nm) assigned v'=15
WIn4902, v'' = 1: excluded midpoint 14.5 from the fit.
```




```
WIn4903: blue peak 43 at 537.0193 nm assigned v'=30
WIn4903: orange peak 15 at 582.4881 nm (nearest 581.278 nm) assigned v'=15
WIn4903, v'' = 1: excluded midpoint 14.5 from the fit.
```




```
WIn6403: blue peak 42 at 536.2590 nm assigned v'=30
WIn6403: orange peak 14 at 582.1005 nm (nearest 581.278 nm) assigned v'=15
WIn6403, v'' = 1: excluded midpoint 14.5 from the fit.
```



	run	sequence	numbering	intercept_cm_inv	u_intercept	slope_cm_inv	u_slope
0	WIn4902	v'' = 0	absolute	127.56888	1.11096	-1.98925	0.01888
1	WIn4902	v'' = 1	absolute	120.28415	2.19907	-1.61980	0.12900
2	WIn4903	v'' = 0	absolute	128.06378	1.09363	-2.01202	0.01789
3	WIn4903	v'' = 1	absolute	118.17650	1.30702	-1.49968	0.06554
4	WIn6403	v'' = 0	absolute	129.73961	0.84888	-2.03187	0.01417
5	WIn6403	v'' = 1	absolute	119.98883	1.21571	-1.59416	0.05575

E Analysis details

Peak indices refer to the zero-based list sorted by increasing reciprocal wavelength, after smoothing with a seven-point, second-order Savitzky–Golay filter. Peak detection used a minimum prominence of 0.01 and a nominal minimum separation of 0.4 nm; detected peaks were restricted to 16 000–20 000 cm^{-1}.

For WIn4903, the selected $v'' = 0$ indices were 31, 33, 35, 37 and every index from 39 to 64 inclusive. Index 43 was assigned $v' = 29$. The $v'' = 1$ indices were 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 20, 22, 24, 26, 28 and 30. Index 15 was assigned $v' = 15$. The spacing at midpoint 14.5 was omitted without renumbering the retained bands. In particular, detected peak 13 was retained in the original selection.

The covariance matrices for the fitted intercept and slope, in the order (C, M), were approximately

$$\mathbf{C}_0 = \begin{pmatrix} 1.15817 & -0.0187290 \\ -0.0187290 & 0.000319999 \end{pmatrix}, \qquad \mathbf{C}_1 = \begin{pmatrix} 1.70826 & -0.0691538 \\ -0.0691538 & 0.00429528 \end{pmatrix}.$$

Each entry has units of cm^{-2}, since the vibrational coordinate is dimensionless. All derived values were evaluated before rounding. The laser position allowance was 0.16354 nm, obtained from the temporal FWHM of 0.176037 s and the reverse scan speed.

The ground-state calculation used the measured anchor at 537.019296 nm. Its covariance with the fitted parameters was retained by propagating directly from the band-position covariance matrix. The additional digits in this appendix identify the calculation inputs, rather than the experimental precision.

F Use of Artificial Intelligence

Artificial intelligence was used to assist with preliminary data analysis to assess scans in the lab, code debugging, and grammatical checking of report text.